

2024



Ferrari Advances Generative AI for Customer Personalization and Production Efficiency

Learn how Ferrari uses generative AI on AWS to enhance the customer and vehicle journeys to increase sales, experimentation, and productivity.

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20% faster
car configurations

60% faster
vehicle simulation

Millions



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Overview

Luxury Italian auto manufacturer [Ferrari S.p.A.](#) has a long tradition and innovation over many decades. As a leader in the industry, Ferrari has tapped into the power of generative artificial intelligence (AI) by turning to Amazon Web Services (AWS).



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Ferrari has applied [generative AI](#) to several use cases, from accelerating the vehicle design process to providing personalized services to its customers. “Every day, we try to find new ways of achieving excellence while improving our customers’ experiences and their connection with our brand,” says Silvia Gabrielli, chief digital and data officer at Ferrari.

Opportunity | Building a Cloud Foundation on AWS with Ferrari

Founded in 1947 for sports car racing, Ferrari epitomizes the power of a lifelong passion and the beauty of limitless human achievement. Ferrari, which has cultivated a loyal fan base, has focused on reaching fans and potential customers by offering unique digital experiences on its website and mobile applications. In 2021, Ferrari selected AWS as its preferred cloud provider to advance its compute, analytics, and storage capabilities. “AWS has been key to our IT transformation,” says Gabrielli. “We started our cloud journey a couple of years ago as a strategic initiative. Today, all our critical workloads are on AWS.”

To free its teams from managing the infrastructure for its applications, Ferrari has invested in fully managed services, such as [AWS Fargate](#), a serverless compute solution for containers. “We’ve reduced the total cost of ownership for our infrastructure from 70 percent to 40 percent,” says Alessio Glorioso, product manager at Ferrari. The company has also seen improvements in the reliability and scalability of its applications. For example, the company can run simulations in its product lifecycle management software 60 percent faster than before.

Ferrari remains committed to furthering its technology road map with [generative AI](#), seeing it as an opportunity to improve the vehicle and customer journeys. “Generative AI is something that every company needs to consider,” says Gabrielli. “It’s a game changer. We can use generative AI to increase our productivity and make it simpler for our fans, dealers, and employees to have the best digital experiences with Ferrari.”

Solution | Greater Speed and Ease of Choice Personalizing a Ferrari Using Amazon Bedrock

To bring the luxury experience to customers around the world, Ferrari developed the car configurator on AWS, giving its customers the ability to personalize their own Ferrari, from wheel selection to paint colors to interior options. “There are millions of possible configurations,” says Simone Candoni, digital experience manager at Ferrari. “We want to help our dealers and customers effectively customize their vehicle to suit their needs.”

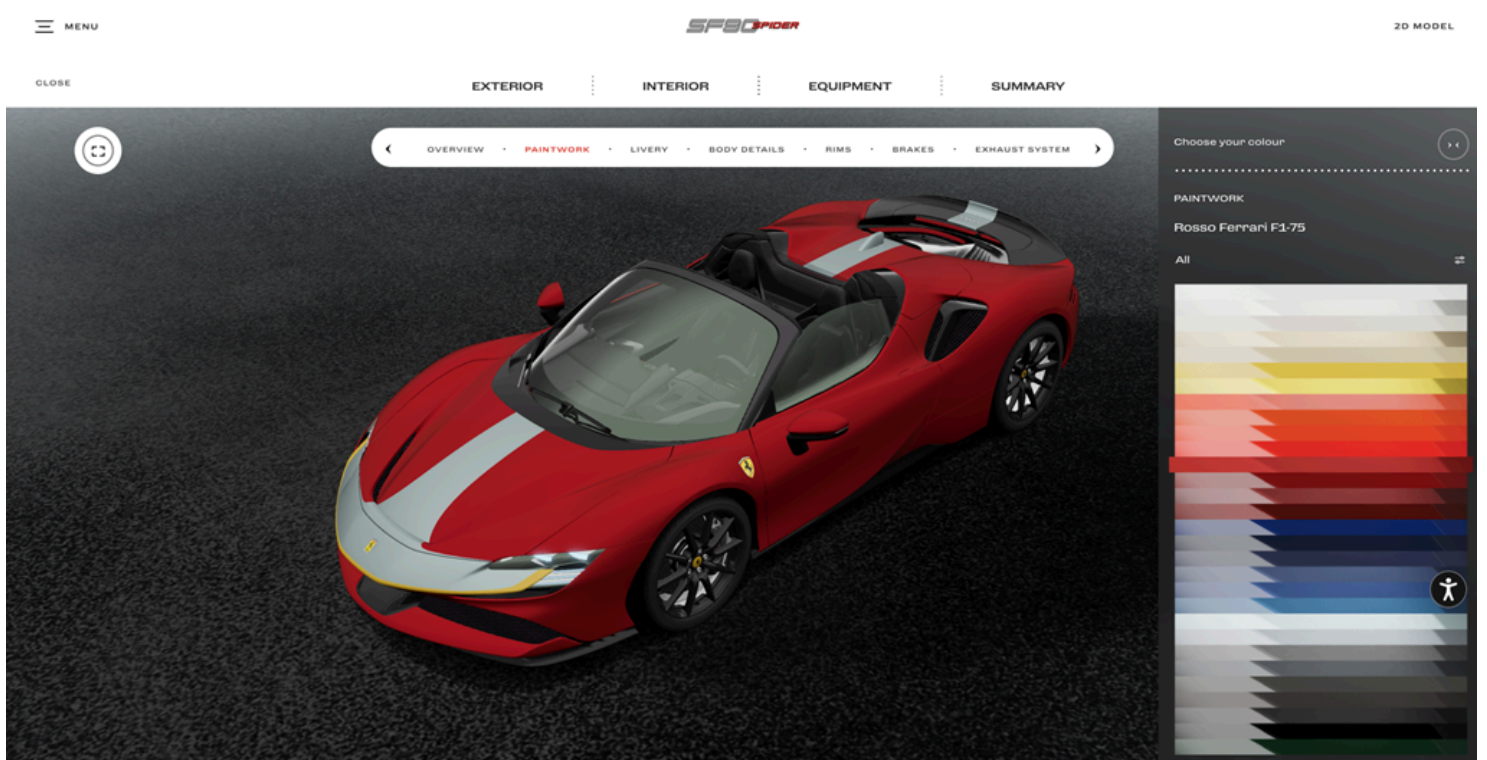
To do so, Ferrari uses large language models (LLMs) in [Amazon Bedrock](#), a fully managed service that offers a choice of high-performing foundation models, along with [Amazon Personalize](#) to elevate the customer experience with personalization powered by machine learning (ML). Since rolling out the car configurator, Ferrari has increased its sales leads and reduced configuration times by 20 percent by offering a more personalized experience where customers can visualize a vehicle with 3D imagery that can be rotated and zoomed in and out. It also allows a customer to virtually experiment with different options.

Ferrari also enhances the after-sales experience with a generative AI chatbot. To assist its sales professionals and technicians, the company fine-tuned LLMs in Amazon Bedrock—including [Amazon Titan](#), Claude 3, and Llama—on its documentation. “Amazon Bedrock has simplified our approach,” says Mauro Coletto, head of business analytics and AI at Ferrari. “We can connect to a single layer of APIs to quickly test, benchmark, and deploy different models.”

Ferrari combines its use of Amazon Bedrock with [Amazon SageMaker JumpStart](#), an ML hub with foundation models, built-in algorithms, and prebuilt ML solutions that can be deployed with only a few clicks. Using these services, Ferrari has trained its chatbot to classify and summarize customer care tickets and answer commonly asked questions that help to reduce human error while improving productivity.

Ferrari uses AI and ML to optimize the production of its vehicles as well. For example, Ferrari uses [Amazon Lookout for Vision](#) to spot product defects using computer vision to automate quality inspections. Using AI, the company can detect missing or defective parts in the assembly line before a vehicle goes to testing, helping it save on costs.

The company has improved the vehicle design with generative AI across F1 vehicles and sports cars by being able to test more designs. Using generative AI, Ferrari can reduce its time to market while relying less on physical prototyping. “We can run analytics and simulations and make correlations that were not possible in the past,” says Gabrielli. “It really speeds up and improves our product development.” Creating full car body physical prototypes is time and resource intensive, and by running virtual simulations on the cloud, Ferrari can run thousands or even millions of simulations in parallel at a very low cost. Additionally, Ferrari is training generative AI text-to-image capabilities that facilitate text-based prompts to make vehicle render improvements and design changes quickly.



Outcome | Experimenting with Generative AI

With a strong cloud foundation on AWS, Ferrari used generative AI solutions to gain measurable impacts across its business. Ferrari will continue expanding its use of generative AI to better serve customers and dealers. The company is also exploring cloud services that will help it meet its carbon neutrality commitment by 2030.

“Innovation is in Ferrari’s DNA, since the first days of our founder,” says Gabrielli. “We see AWS as a strategic collaborator in our innovation efforts. We really align in terms of our values around excellence, innovation, and customer obsession.”

About Ferrari

Italian luxury sports car manufacturer Ferrari has built a legacy upon decades of sporting success, craftsmanship and innovation. Ferrari continues to uphold its tradition of racing with its professional racing team, Scuderia Ferrari.

AWS Services Used

Amazon Bedrock

Amazon Bedrock is a fully managed service that offers a choice of high-performing foundation models (FMs) from leading AI companies like AI21 Labs, Anthropic, Cohere, Meta, Mistral AI, Stability AI, and Amazon through a single API, along with a broad set of capabilities you need to build generative AI applications with security, privacy, and responsible AI.

[Learn more »](#)

Amazon Personalize

Amazon Personalize accelerates your digital transformation with ML, making it easier to integrate personalized recommendations into existing websites, applications, email marketing systems, and more.

[Learn more »](#)

Amazon SageMaker Jumpstart

Amazon SageMaker JumpStart is a machine learning (ML) hub that can help you accelerate your ML journey. With SageMaker JumpStart, you can evaluate, compare, and select FMs quickly based on pre-defined quality and responsibility metrics to perform tasks like article summarization and image generation.

[Learn more »](#)

Amazon Lookout for Vision

Amazon Lookout for Vision is an ML service that uses computer vision to spot defects in manufactured products at scale.

[Learn more »](#)



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